O list E-Commerce Seller Performance Analysis

Dataset: Olist Brazilian E-Commerce Dataset
Tools: MySQL Workbench, SQL, Microsoft Excel

Executive Summary

This project evaluates seller performance in the Olist e-commerce marketplace using five analytical views: revenue, order volume, average delivery time, delivery delay and cancellation rates, and average freight cost.

The revenue chart shows that the highest revenue seller generated 917.80, followed by sellers generating 810.00 and 809.10. The order-volume chart shows that the highest-volume seller handled nine orders, while the other nine sellers handled between three and seven orders each.

The delivery-time analysis identifies ten sellers with average delivery times ranging from 36 to 68 days. The seller with the longest average delivery time recorded 68 days, indicating an area that may require further investigation.

The delay and cancellation chart shows that one of the five sellers analyzed had a delay rate of 14.29%, while the other four had delay rates of 0%. All five had cancellation rates of 0% in the results displayed.

The freight comparison shows an average freight cost of 10.83 for the best-performing ten sellers and 27.74 for the worst-performing ten sellers. The worst-performing group therefore had an average freight cost approximately 2.56 times higher.

Overall, the findings demonstrate the importance of evaluating sellers using both commercial and operational measures. However, some of the seller-level samples are small, and the observed differences do not establish that any single factor causes weaker performance.

Business Problem and Objectives

E-commerce businesses rely on sellers to generate revenue, process orders efficiently, and deliver products within expected timeframes. Differences in seller performance can affect operational efficiency and customer experience.

The main business question was:

How does seller performance vary across sales, order fulfillment, delivery reliability, and freight costs, and what actions could improve operations?

The project objectives were to:

- Identify sellers generating the highest revenue.
- Compare order volumes across sellers.
- Identify sellers with long average delivery times.
- Examine delivery delay and cancellation rates.
- Compare average freight costs between stronger- and weaker performing seller groups.
- Recommend actions to improve seller monitoring and fulfillment efficiency.

Data and Methodology

The analysis used SQL queries in MySQL Workbench to aggregate order and seller information from the Olist dataset. The query results were exported to Excel, where charts were created to support business interpretation.

The main indicators were:

Indicator	Purpose
Revenue	Measure sales value associated with each seller
Order volume	Measures the number of distinct orders associated with each seller
Average delivery time	Measures elapsed days between purchase and customer delivery
Delay rate	Measures the percentage of delivered orders received after the estimated date
Cancellation rate	Measures the percentage of seller-associated orders marked as cancelled
Average freight cost	Compare freight costs between selected seller groups

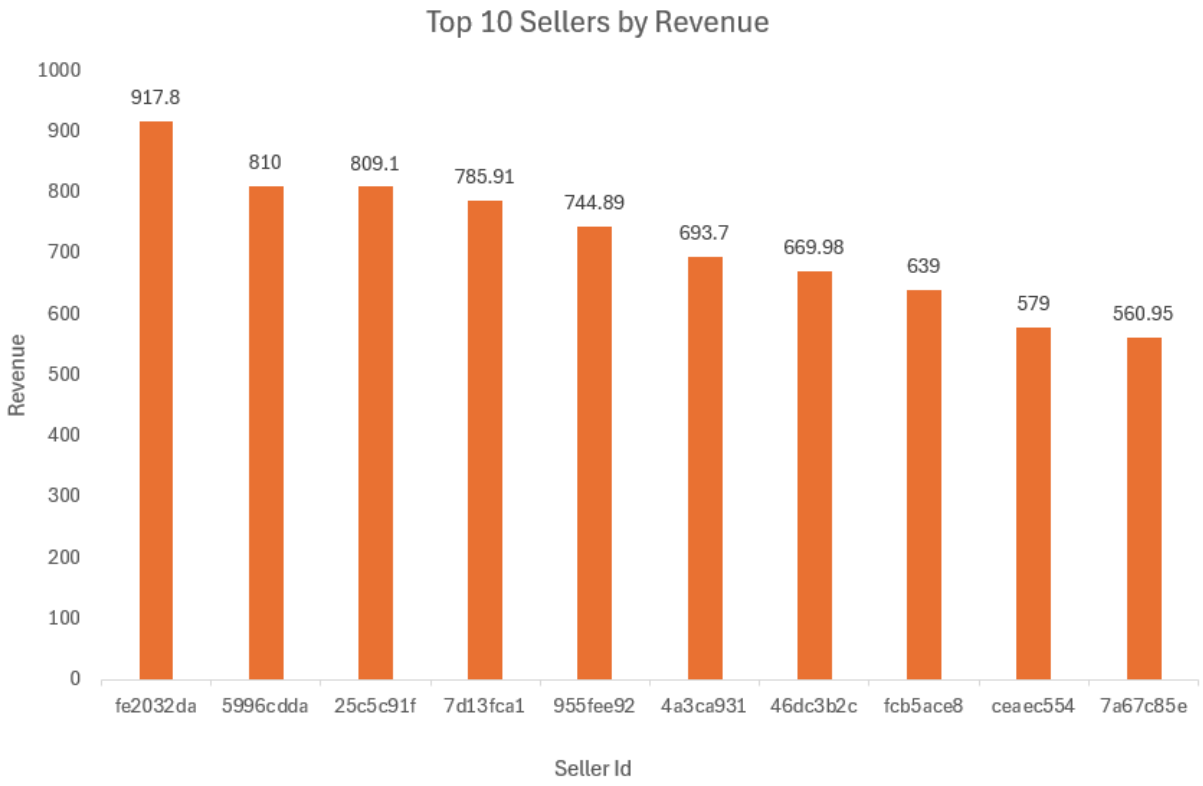
Revenue and freight values are reported in the units returned by the SQL queries. The freight comparison is not assumed to measure profitability, and the analysis does not establish causal relationships between the indicators.

Findings and Chart Analysis

1. Revenue Analysis

Top 10 Sellers by Revenue

The revenue chart shows the following ten sellers, ordered from highest to lowest revenue:



The highest revenue seller generated **917.80**, while the tenth seller generated **560.95**. The difference between these values is **356.85**.

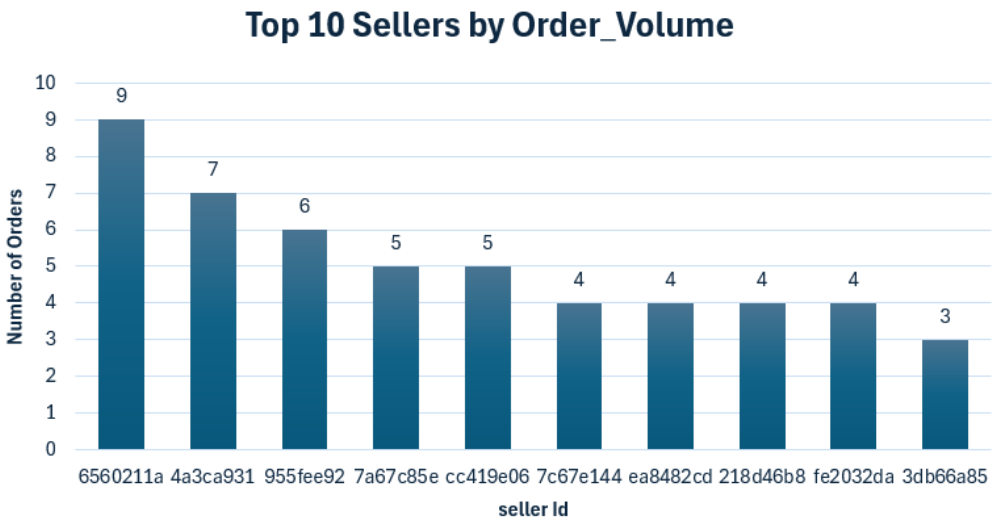
These results show that sales contribution varies between the ten sellers. High-revenue sellers may be important to the marketplace's commercial performance, but revenue alone does not show whether their deliveries are timely, or their freight costs are low.

Business implication: Monitor revenue alongside fulfillment indicators so that commercially important sellers can be evaluated for both sales contribution and service reliability.

2. Order Volume Analysis

Top 10 Sellers by Order Volume

The chart shows the following order counts:



The highest-volume seller handled nine orders, while the tenth seller handled three. The ten sellers collectively account for 51 seller-associated orders as displayed in the chart.

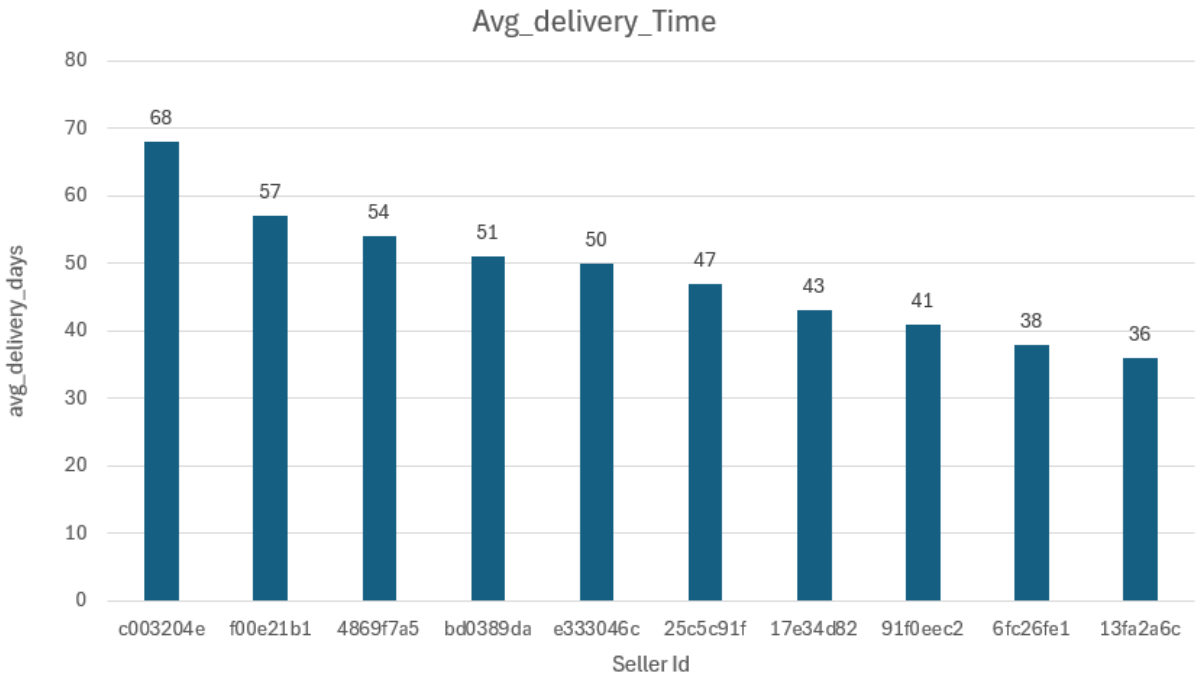
The seller with the highest order volume is not the same seller as the highest revenue seller. This illustrates why order volume, and revenue should be analyzed separately: sellers may handle different numbers of orders and generate different amounts of revenue.

Business implication: Consider both order volume and revenue when identifying important sellers. Further analysis using a larger dataset would help determine whether the differences persist over time.

3. Average Delivery Time Analysis

Average Delivery Time by Seller

The ten sellers in the delivery-time chart have average delivery times of 68, 57, 54, 51, 50, 47, 43, 41, 38, and 36 days.



The longest average delivery time was 68 days, while the shortest among these ten sellers was 36 days. The difference between the two is 32 days.

This chart identifies sellers whose delivery times may warrant investigation. However, the results do not establish why these deliveries took longer. Possible factors to investigate include order processing time, shipping distance, carrier performance, product characteristics, and logistics arrangements.

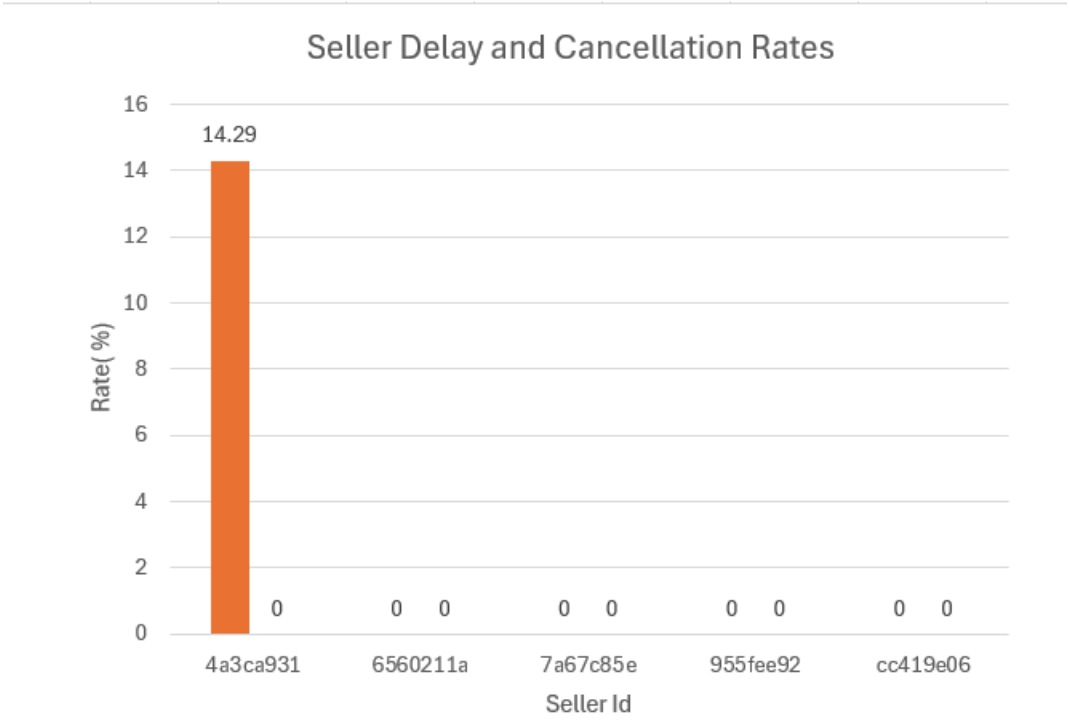
The average delivery-time measure should also be interpreted alongside the estimated delivery date. A long delivery time is not necessarily a late delivery if the estimated delivery window was also long.

Business implication: Investigate sellers with long average delivery times and compare their actual delivery dates with estimated delivery dates. This can help distinguish long but expected deliveries from deliveries that fail to meet customer expectations.

4. Delivery Delay and Cancellation Analysis

Seller Delay and Cancellation Rates

The completed chart includes five sellers. Their results were:



One seller recorded a delay rate of 14.29%, while the other four recorded 0%. All five had cancellation rates of 0% in this result set.

The 14.29% delay rate represents one late delivery among seven delivered orders. Because the sellers in this chart have relatively few orders, one additional late delivery could substantially change a seller's percentage.

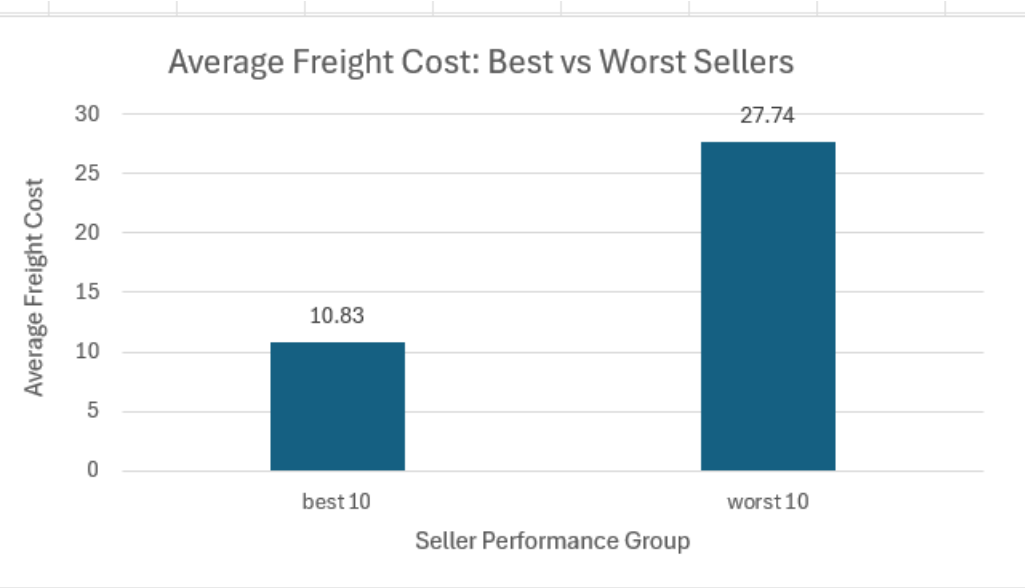
These results describe only the five sellers shown and should not be interpreted as the overall delay or cancellation rates for the marketplace.

Business implication: Continue monitoring delivery delays and cancellations using a larger seller sample and a consistent reporting period. Investigate repeated problems rather than drawing strong conclusions from a small number of orders.

5. Freight Cost Comparison

Average Freight Cost — Best vs. Worst Sellers

The comparison produced the following values:



The average freight cost for the worst performing group was approximately 2.56 times the average for the best performing group. The absolute difference was 16.91.

This is a substantial difference between the two selected groups. It suggests that freight cost is worth investigating as part of the broader seller-performance analysis.

However, the chart does not establish that higher freight costs cause poor seller performance. Shipping distance, package size, product mix, delivery destination, and carrier arrangements may influence freight costs. These factors should be investigated before making operational changes.

Business implication: Review freight costs by seller, destination, product characteristics, and carrier. Evaluate whether cost-saving opportunities exist without reducing delivery reliability or customer service.

Overall Business Interpretation

The five charts show why seller performance should be assessed using multiple indicators.

Revenue identifies sellers with higher sales contributions, while order volume measures the number of orders they handle. Average delivery time highlights differences in delivery speed, and delay and cancellation rates provide additional information about fulfillment

reliability. Freight costs offer another perspective on the operational differences between seller groups.

The analysis reveals three particularly relevant observations:

1. **Commercial performance differs across sellers.** The highest revenue seller generated 917.80, while the highest volume seller handled nine orders. Revenue and order volume therefore provide distinct information.
2. **Some sellers have long average delivery times.** The ten sellers in the delivery-time chart ranged from 36 to 68 days, warranting further investigation into delivery processes and estimated delivery dates.
3. **Freight costs differ between the selected performance groups.** The worst-performing group had an average freight cost of 27.74, compared with 10.83 for the best-performing group.

The delay and cancellation results should be interpreted cautiously because the chart contains only five sellers, each with relatively few orders. Similarly, the freight comparison indicates a difference between groups but does not identify the underlying cause.

Recommendations

Develop a seller performance scorecard

Create a standardized scorecard that includes revenue, order volume, average delivery time, delay rate, cancellation rate, and average freight cost. This would provide a balanced view of seller performance.

Investigate long delivery times

Review sellers with the longest average delivery times. Examine processing time, dispatch delays, delivery distance, carrier performance, and estimated delivery dates to identify potential causes.

Monitor delivery reliability over larger samples

Use consistent reporting periods and sufficient order volumes when comparing delay and cancellation rates. Small samples should be treated as preliminary signals rather than definitive assessments.

Investigate freight-cost differences

Determine why the worst-performing group has a higher average freight cost. Compare destinations, shipping distances, product characteristics, and carrier arrangements before recommending changes.

Evaluate sellers using both revenue and operations

Avoid evaluating sellers based solely on sales contribution. Consider whether sellers with high revenue or high order volume are also meeting delivery expectations.

Review performance regularly

Produce monthly or quarterly reports to identify changes in revenue, delivery performance, cancellations, and freight costs. Repeated analysis would help determine whether operational improvements are effective.

Limitations

The findings should be interpreted while considering the following limitations:

- The revenue chart shows the top ten sellers and does not, by itself, establish overall marketplace revenue distribution.
- The order-volume chart contains small order counts, which may not represent long-term seller activity.
- The delivery-time chart identifies sellers with long average delivery times but does not establish the causes of those times.
- The delay and cancellation chart contains only five sellers and should not be generalized to all sellers.
- Orders associated with multiple sellers can appear in the analysis for each associated seller.
- The freight-cost comparison does not establish causation and does not measure profitability.
- The charts do not control for all factors that could affect seller performance.

Future analysis could include a larger sample, delivery distance, product category, carrier information, processing time, and a consistent observation period.

Conclusion

This project demonstrates how SQL and Excel can be used to analyze e-commerce seller performance across commercial and operational indicators.

The revenue and order-volume charts identify differences in seller contribution. The delivery-time chart highlights sellers whose fulfillment speed warrants investigation, while the delay and cancellation chart shows how seller-level reliability measures can vary. The freight comparison identifies a substantial cost difference between the selected best- and worst performing groups.

The most important next step is to bring these measures together in a consistent seller performance scorecard and investigate the operational factors behind weaker results. This would help the business make better-informed decisions about seller support, delivery reliability, and freight-cost management.

Evaluating revenue, order volume, delivery time, delay rates, cancellations, and freight costs together provides a more useful understanding of seller's performance than relying on sales figures alone.